10	**(a)**		[=]153	**B1** **[1]**	**1.1**		
10	**(b)**		$z = \pm 1.645$ used their positive 1.645 = $\frac{183 - their\ 153}{\sigma}$ oe (= 18.237…to 18.248…) σ = 18.2 cao	**B1** **M1** **A1** **[3]**	**1.1** **2.1** **1.1**	or ±1.644(85…)	**M0** if continuity correction used

Question			Answer	Marks	AOs		Guidance
10	(c)		$\left[\frac{16}{452} =\right] 0.035(398\ldots)$	**B1**	**3.1b**	allow percentage	or **B1** as main scheme then Invnorm(0.0353, their 153, their 18.2) for **M1**
			their P($X < 120$) from N(their 153, their 18.2^2)	**M1**	**1.1**	**M0** if continuity correction used	**NB** 119.95 to 120.15 which is close to 120 oe for **A1**
			probability of 0.0349 to 0.0352 which agrees to 2 sf	**A1**	**2.2b**	allow percentage	
				[3]		…… **or B1** for their P($X < 120$) then **M1** for 452 × their 0.03490.. then **A1** for 15.77 to 15.91 which is close to 16 oe …… **or B1** as main scheme then **M1** for $\sigma = \frac{120-153}{their\ -1.809}$ and 18.3 ≈ 18.2 for **A1**	**or B1** as main scheme then $z = \frac{120 - their\ 153}{their\ 18.2}$ for **M1** **NB** –1.809 to –1.813 and Invnorm(0.0353, 0, 1) to obtain –1.806 to – 1.812 which is close to –1.809 to –1.813) for **A1**

Question			Answer	Marks	AOs		Guidance
10	(d)		H_0: μ = their 153 H_1: μ < their 153	**B1**	**1.1**	for both hypotheses; may be stated in words, but need to see 153 for **B1** and **population** mean for 2nd **B1**	**B0** if other parameter used unless clearly defined as population mean
			μ is the **population mean** flight time from **Magaluf to Liverpool**	**B1**	**2.5**		
			use of N(their153, $\frac{their18.2^2}{24}$) to find $P(\bar{X} < 143.6)$	**M1***	**3.3**	or inv Norm(0.01, their 153, $\frac{their\ 18.2^2}{24}$)	or $z = \frac{143.6-their153}{their\frac{18.2}{\sqrt{24}}}$
			awrt 0.0057 to 0.0058… to 2 or more sf oe **isw**	**A1**	**1.1**	$\bar{X}$ < 144 to 144.4 is critical region	z = – 2.5248.. to –2.5302… to 2 or more dp
			their 0.0057 correctly compared with 0.01 oe	**M1dep***	**3.4**	143.6 correctly compared with their 144.36 FT their comparison	their z compared with – 2.326 or – 2.33
			result is significant or reject H_0 or accept H_1	**A1**	**1.1**		
			there is sufficient evidence to **suggest** at the 1% level that the **mean** flight time from **Magaluf to Liverpool** is less than **153** / mean flight time from Liverpool to Magaluf	**A1**	**2.2b**	FT their comparison	do not allow eg conclude / prove / indicate or other assertive statement instead of suggest
				[7]			

Question			Answer	Marks	AOs		Guidance
10	(e)		**reduce** the value of μ **increase** the value of σ or σ^2	**B1** **B1** **[2]**	**3.5c** **3.5c**	allow **B1** for eg use new sample data to calculate new estimate for σ or σ^2	eg take a bigger sample is insufficient